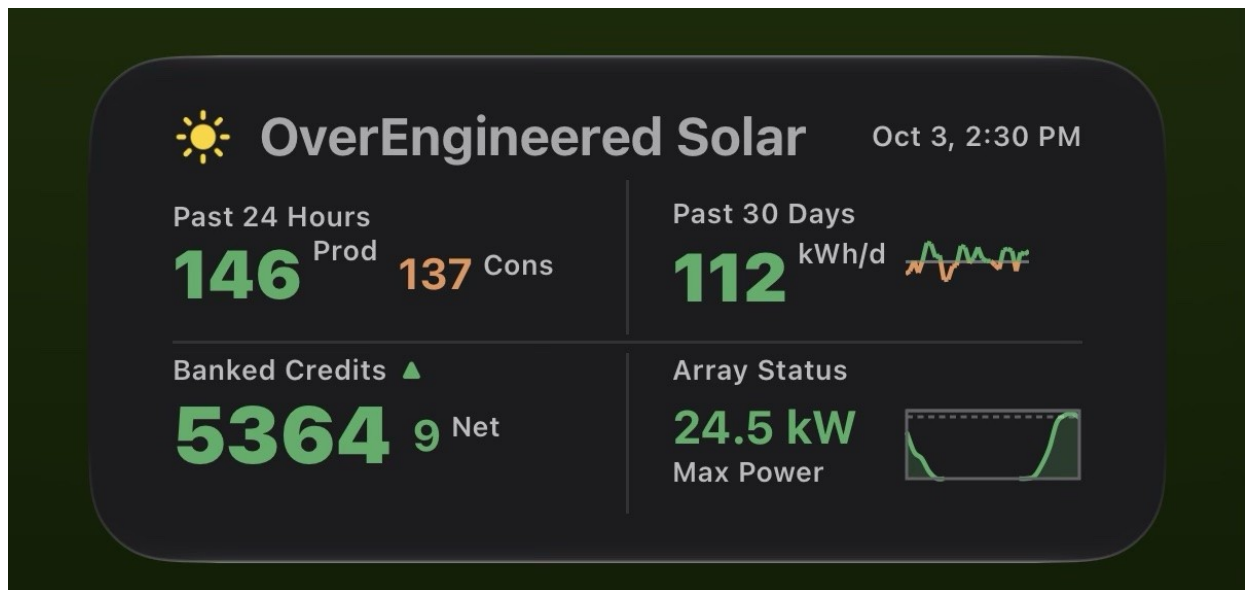


OverEngineered Solar

A home-screen widget for the Scriptable app on iPhone, for use with Enphase® solar systems.

Guide version 2026.10.05.2013. Written for widget 2026.10.05.1740 and setup 2026.10.05.1740.

Everything here runs on your iPhone, inside the Scriptable app. A computer is only useful for reading this guide and, if you like, creating the Enphase developer account.



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What it shows

Title row. A weather icon for your location, your title, and the time of the newest data, which tells you how fresh the numbers are. The time turns orange if the last update failed.

Past 24 Hours. Production and consumption over the last 24 hours, in kWh. This is a rolling window ending now, not “since midnight”, so on two similar days the numbers stay steady.

- **Prod** is green when it is at or above your 30-day average production, orange when below.
- **Cons** is green when it is at or below your 30-day average consumption, orange when above.

Past 30 Days. Your average daily production over the last 30 days, with a 30-day trend line. Green means you are producing at least as much as you use, on average.

Banked Credits. Your running net-metering balance in kWh: the balance from your last bill, plus everything produced minus everything consumed since then.

- The small mark after the label shows which way the bank has been heading over the last 28 days: a green up triangle (adding more than about 5 kWh a day), an orange down triangle (drawing down more than about 5 kWh a day), or a grey dash (roughly flat).
- **Net** is production minus consumption for the same rolling 24 hours shown at the top left.
- If the estimate goes below zero the label reads “Credit Deficit”.

Array Status. A health check and a chart.

- The chart is the last 24 hours of production. The right end is your current power. The dashed line marks the highest reading in the last hour.
- The number is your array's peak power across the last five clear days. The label under it is a verdict:
 - **Max Power** — normal.
 - **Dirty** or **Very Dirty** — clear-day peaks have dropped; the panels may need washing.
 - **Winter Sun** — the sun is too low to judge, so the number holds its last real value until spring.
 - **Service Req** — Enphase is reporting a hardware fault.

Tap the widget to open the Enphase app (see “The tap shortcut” below).

What you need

- An iPhone with the free Scriptable app.
- An Enphase system and the login you use in the Enphase app.
- A free Enphase developer account. Setup walks you through creating one.

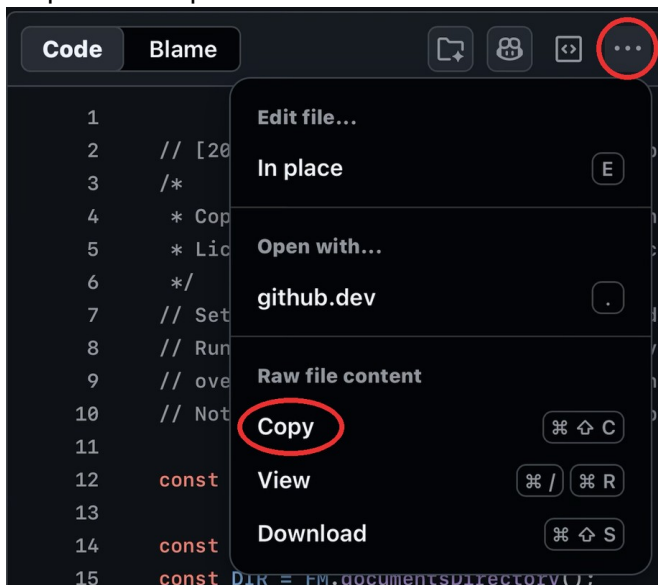
Consumption monitoring is optional. Without it, setup asks for a daily usage estimate.

Install

Scriptable is a free iPhone app that runs small scripts and can show one as a home-screen widget. This widget is two of those scripts: one you run once to set things up, and one that draws the widget.

Do all of this on your iPhone. The scripts have to be in Scriptable on the phone, and setup has to run there, because that is where it reads your keys and saves your settings.

1. Install **Scriptable** from the App Store.
2. In Safari on your iPhone, open this project's GitHub page and tap `setup-overengineered-solar.js` in the file list.
3. Tap the **⋮** at the right of the row that says **Code | Blame**, then tap **Copy** under "Raw file content". The whole script is now copied.



4. Open Scriptable and tap **+** at the top right. Paste, then tap **Done** at the top left.
5. Back in the script list, press and hold the new **Untitled Script**, tap **Rename**, and call it **OverEngineered Solar Setup**.
6. Go back to GitHub and do the same with `widget-overengineered-solar.js`: copy it, paste it into a new script, tap **Done**, then rename that one **OverEngineered Solar**.
7. In Scriptable, tap **OverEngineered Solar Setup** to run it (next section).
8. When setup is finished, add a medium Scriptable widget to your home screen, long-press it, choose **Edit Widget**, and pick **OverEngineered Solar** as the script.

The script names are suggestions. Any names work, as long as you can tell the two apart.

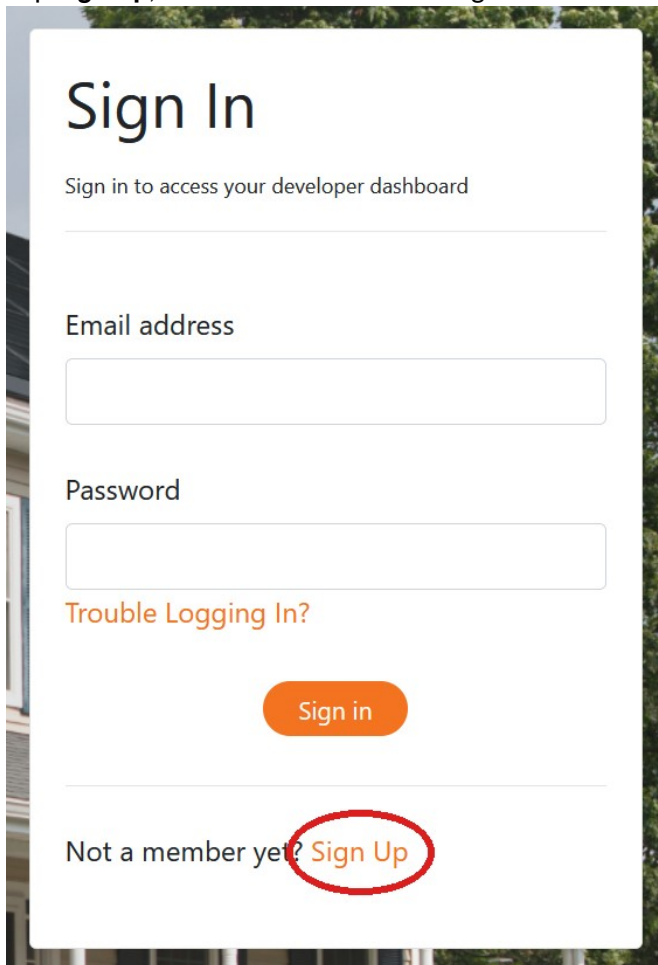
Setup

In Scriptable, tap **OverEngineered Solar Setup** in the script list to run it. The first run has three parts. Everything happens inside Scriptable; whenever you are on an Enphase page, the **✕ at the top left** brings you back.

Tip: print this guide, or keep it open on a second screen, so the steps and pictures are beside you while you work on your phone.

Part 1 — Get your keys

1. Tap **Open Enphase Developer Site**. If a cookie box appears, tap **Accept All Cookies**.
2. Tap **Sign Up**, the small link under the Sign In button.



Sign In

Sign in to access your developer dashboard

Email address

Password

[Trouble Logging In?](#)

[Sign in](#)

Not a member yet? [Sign Up](#)

3. Fill in the form. **Organization/Group Name** and **Username** can be anything, such as “Home”.

Sign Up

Organization/Group Name *

Self

Username *

MyEnphaseApp

Email *

you@example.com

Password *

.....

Password confirmation *

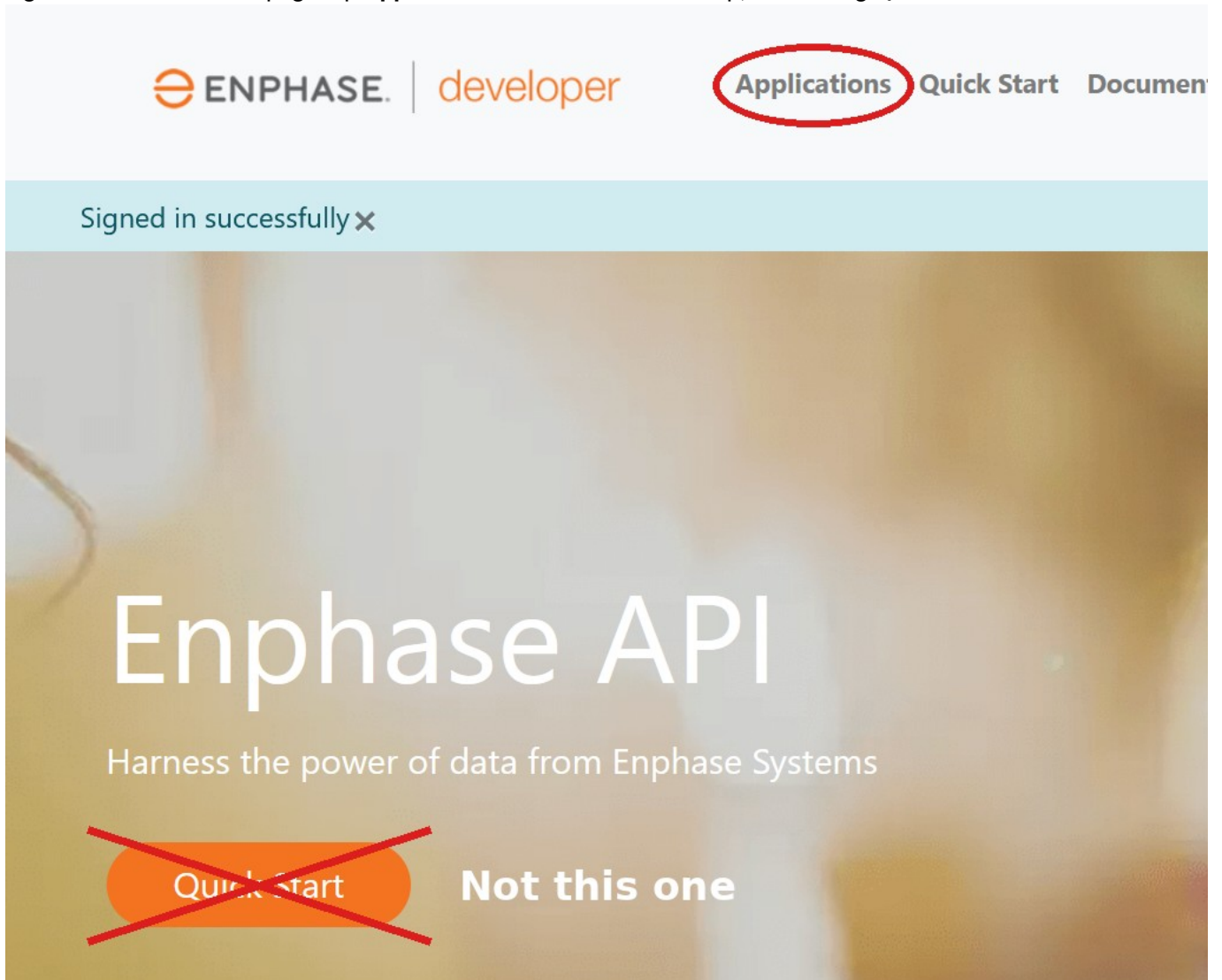
.....

☒ I have read the [Privacy Policy](#) and by signing up, I agree to the [API License Agreement](#), and [Terms of Service](#).

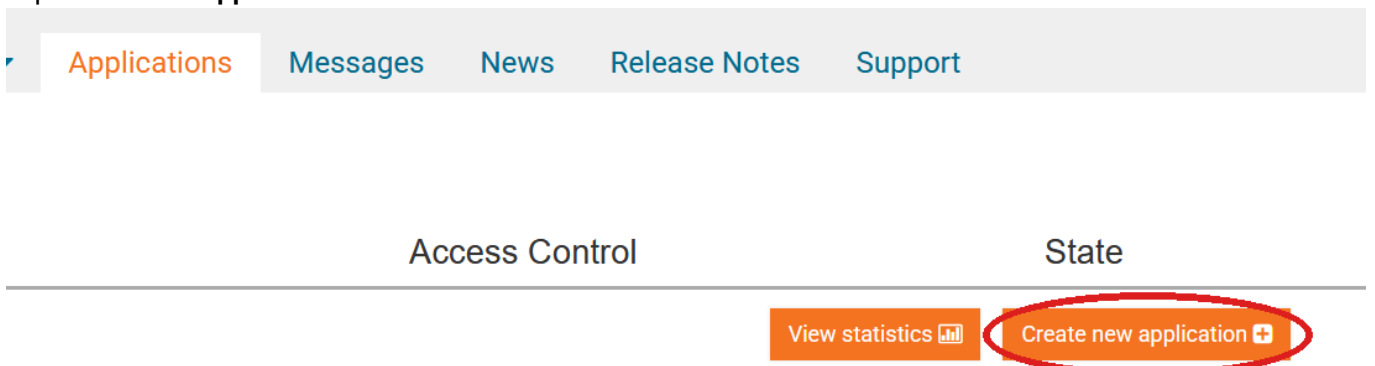
Sign Up

4. Enphase emails you an activation link. It comes from api@enphaseenergy.com with a subject starting “Action Required”; check spam if you don't see it. Tap the link. It opens a sign-in page. (A second tap says “already activated”, which is fine.)

5. Sign in. On the welcome page tap **Applications** in the menu at the top, not the big Quick Start button.



6. Tap **Create new application**.



7. Leave the plan on **Watt** (free). Name, Description and Developed By can be anything.

8. **Access Control: check every box.** They all start unchecked, and the widget can't read your data without them.

Select Plan *

Watt

▼

[Plan Details](#)

Name *

MyReallyCoolSolarWidget

Description *

Enphase Solar Compatible Scriptable iOS Widget

Developed By *

OverEngineered Tech

Access Control *

Select the access controls required for your application from the list below

☒ System Details

☒ Site Level Production Monitoring

☒ Site Level Consumption Monitoring

☒ EV Charger Monitoring

Create Application

9. Tap **Create Application**. After a moment the page reloads and shows your **API Key**, **Client ID** and **Client Secret**. Wait until you can see all three.

Name	MyReallyCoolSolarWidget		
Description	Enphase Solar Compatible Scriptable iOS Widget		
Plan	Watt › Review/Change › Add Credit Card Details		
Status	Live		
API Key		Add this as a key parameter to your AF	
	Regenerate		
Access Control	System Details Site Level Production Monitoring Site Level Consumption Monitoring EV Charger Monitoring		
Client ID			
Client Secret			
Developed By	OverEngineered Tech		
Authorization URL			

10. Tap the ✕ at the top left. Setup reads the three keys for you. (You can also type them in.)

Optional: the Enphase developer site is easier on a bigger screen. You can create the developer account and the application on a computer if you prefer. You still run setup on your iPhone afterwards and sign in there, and it reads the keys the same way.

Part 2 — Approve access

1. Tap **Continue to Enphase login**.
2. Log in with your normal Enphase account, the one you use in the Enphase app. This is not the developer account.
3. Tap **Authorize**, then the ✕.

This lets the widget read your system. It cannot change anything.

Part 3 — Fill in the form

Your system ID is filled in for you. Enter the rest, tap **Save**, then the ✕.

Setting	What to enter
System turn-on date	The day your system started producing. Data is counted from this day; pick a later date if your early data was wrong.
Peak AC power	The highest kW you have seen in the Enphase app.
Panel tilt	In degrees. Use 30 if unsure.
ZIP code	For weather, sunrise and sunset.
Banked credits	The kWh balance and date from your most recent bill, or 0.
True-up month	The month your utility resets banked credits.
Enphase shows consumption	On if the Enphase app shows consumption, not just production.
Title	The name shown on the widget.

Changing a setting later

Run setup again any time. It opens straight to the form, with no login. The usual reason is a new banked-credit balance from a bill.

The tap shortcut

This is optional. It makes a tap on the widget open the Enphase app.

In Apple's **Shortcuts** app, make a new shortcut with one action, **Open App**, set to the Enphase app (listed as **Enlighten**). Name the shortcut exactly `Open Enlighten`.

On newer iPhones you can type “Open the Enlighten app” in the **Describe a shortcut** box. The look of the Shortcuts app varies between iOS versions.

Updating to a newer version

Paste the new code over the old script in Scriptable. Your settings and login are kept in separate files, so there is nothing to redo.

Messages you might see

On the widget	Meaning
Run the setup script first	No settings were found. Run setup.
Reauth needed	Enphase no longer accepts the saved login. Run setup and tap “Log in again”.

On the widget	Meaning
Something went wrong	An unexpected error. Run the widget script inside Scriptable to see the details.

Why it isn't fully real time

The widget updates about once an hour while the sun is up (occasionally an hour and a quarter), every one to two hours in the evening, and not at all between 10 PM and 5 AM. The time in the top right corner is the time of the newest data, so it shows how fresh the numbers are. That pace is on purpose, and there are two reasons.

- **Enphase's free plan allows 1,000 requests a month.** That is roughly 33 a day, and one refresh can take more than one request. Updating every few minutes would use up the month in a few days.
- **iOS decides when a widget runs.** A script can ask to be woken, but iOS picks the actual moment and often runs it late. No home-screen widget can be truly live.

One more thing shapes the numbers: Enphase publishes each 15-minute reading before it is complete, and takes about ten minutes to finish it. The widget waits for the finished reading before showing it as “current power”, so that figure is typically 10 to 25 minutes behind the Enphase app.

How it stays under 1,000 requests

For anyone who wants the detail, including anyone at Enphase:

- The widget counts every request it makes and keeps the running total for the month.
- It plans each month to about 950 requests and refuses to go past 990, leaving a margin under the limit.
- Nothing is requested between 10 PM and 5 AM.
- Production is requested about hourly while the sun is up.
- Consumption is requested every two hours, and hourly near the start and end of the day when the month has requests to spare.
- Daily totals and the fault check are requested once a day.
- Opening the script by hand only uses spare requests. If there are none, it shows saved data.

In simulated months, including heavy-use ones, the busiest came to about 950 requests and no day went without data.

Privacy

Everything stays on your phone. Setup saves two files in Scriptable's local storage:

- `overengineered-solar-config.json` — your settings and keys.
- `overengineered-solar-token.json` — your Enphase login tokens.

The scripts talk only to Enphase, to Open-Meteo for weather, and to Zippopotam.us to turn your ZIP code into coordinates.

Debugging

Run the widget script inside Scriptable. It shows a status report and copies it to the clipboard. Keys and tokens appear only as their last four characters. The setup script copies a similar report when it finishes.

Support

This widget is free and always will be for personal use. If it's useful to you and you'd like to say thanks, you can buy me a coffee: buymeacoffee.com/overengineeredtech

